Power Budget

Team Number: 209
Project Name: **Automated Laundry Folding Robot**
Team Member Names: **Diego, Ruben, Kristen, Zainan**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Gearmotor	FIT0186	12 V	1	7000	7000	mA
	H-Bridge	FAN8100N	2.2V ~ 9V	1	3000	3000	mA
	Curiosity Nano	PIC18F57Q43	1.8V ~ 5.5V	1	100	100	mA

+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Gearmotor	FIT0186	12 V	1	7000	7000	mA
	H-Bridge	FAN8100N	2.2V ~ 9V	1	3000	3000	mA
					Subtotal	10000	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	12500	mA
Source	12V Wall Supply	GST60A12-P1J	12V	1	5000	5000	mA
					Total Remaining Current Available on +12V Rail	-7500	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Curiosity Nano	PIC18F57Q43	1.8V ~ 5.5V	1	100	100	mA
							mA
					Subtotal	100	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	125	mA
Regulator	+5V Regulator	LM7805	7V ~ 35V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	1375	mA

Unused Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	0	mA
Regulator	N/A	N/A	N/A	0	0	0	mA
					Total Remaining Current Available on 3.3V Rail	0	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	12V Wall Supply	GST60A12-P1J	100-240 VAC	12V	5000	5000	mA
Power Rails Connected to External Power Source 1	+12V Motor Rail	GST60A12-P1J	+12V - 35V	1	10000	10000	mA
	+5V Regulator	L7805CV	(range)	1	100	100	mA
	Unused rail	N/A	N/A	0	0	0	mA
				Total Remaining Current Available on External Power Source 1		-5100	mA

External Power Source 2	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	N/A	N/A	N/A	N/A	0	0	mA
Power Rails Connected to External Power Source 2	N/A	N/A	N/A	0	0	0	mA
				Total Remaining Current Available on External Power Source 2		0	mA

Calculated Battery Life (N/A)							
	Component Name	**Part Number**	**Supply Voltage Range**		**Capacity (mAh)**	**Required By Regulators**	
	N/A	N/A	N/A		0	0	
					Battery Life	0	hours

Notes:

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit